

Lab Search Sample Paper

This paper studies model merging for efficient reuse of language models.

Model merging combines parameters from independently trained expert models.

Parameter interpolation is a simple baseline that averages model weights.

Task arithmetic can add or subtract learned parameter update vectors.

The evaluation compares merged models on accuracy and robustness benchmarks.

Good merge coefficients preserve useful capabilities from every source model.

Our retrieval prototype extracts text from PDF documents before chunking.

Each chunk receives a deterministic local hashing embedding for development.

The backend computes cosine similarity between a query and indexed chunks.

Results are sorted by similarity score and the top ranked chunks are returned.

The web application sends requests to the backend REST API over HTTP.

A future Slack bot will call the same API and will contain no search logic.

The repository abstraction can later use Supabase and pgvector search.

The embedding provider can later be replaced by Gemini or an E5 model.

This Phase One system intentionally omits answer generation and reranking.